

IMDB Sentiment Analysis Report

Assignment: Individual project - large language models (25%)

Module: Research Methods in Data Science

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GitHub Repository: <https://github.com/deepakraj-04/Individual-Project-Large-Language-Models>

Introduction:

Sentiment analysis is an NLP (Natural Language Processing) problem where texts are classified into positive or negative opinions. Sentiment analysis is used to examine customer reviews, social media posts and product feedback which helps in organizations to make business decisions.

This assignment performs sentiment analysis on IMDB movie review dataset using two different methods. We apply TF-IDF as feature extraction with Logistic Regression algorithm. Then we use pre-trained bert-base-uncased model which is fine-tuned on IMDB dataset for binary sentiment classification.

IMDB dataset consisting of 4000 reviews for training, 1000 reviews for validation and 2000 reviews for testing. The performance with both models was evaluated based on Accuracy, Precision, Recall and F1-score. The aim of the assignment is to compare the performance of a machine learning model against Fine-Tuned Transformer model for sentiment analysis.

Methodology:

The IMDB movie review dataset was used in this assignment. The dataset consists of 4000 training reviews, 1000 validation reviews and 2000 test reviews was selected to save time and still have an equal amount of positive and negative sentiment classes.

The reviews for the Logistic Regression model were converted to numeric features using Term Frequency-Inverse Document Frequency (TF-IDF) method. The TF-IDF method determines the importance of each word in the review and reduces the effect of commonly occurring words. The features were used to train the Logistic Regression.

The second model is based on a pre-trained bert-base-uncased transformer model. The review text was tokenized using the BERT Tokenizer with a maximum number of tokens was set to 256. Fine-tuning of the pre-trained model was done on the IMDB dataset with the Hugging Face Trainer.

The training of both models was performed on the same dataset split. Logistic Regression model used TF-IDF features to represent the text numerically. The BERT model processed the review text using the tokenizer before fine-tuning the model. Optimization, validation, and evaluation of the models were done with the help of Hugging Face Trainer and the model was selected based on the highest validation F1-score.

Both models were tested on the test dataset using Accuracy, Precision, Recall, and F1-score metrics. These metrics was used to compare the performance of the Logistic regression model and Fine-Tuned Transformer model.

Results and Discussion:

Both Logistic regression model and Fine-Tuned Transformer model have shown excellent classification accuracy on the data set. Logistic Regression model has achieved an accuracy of 85.05% and an F1-score of 85.36% while the Fine-Tuned Transformer model has produced accuracy of 90.95% and F1-score of 90.99%.

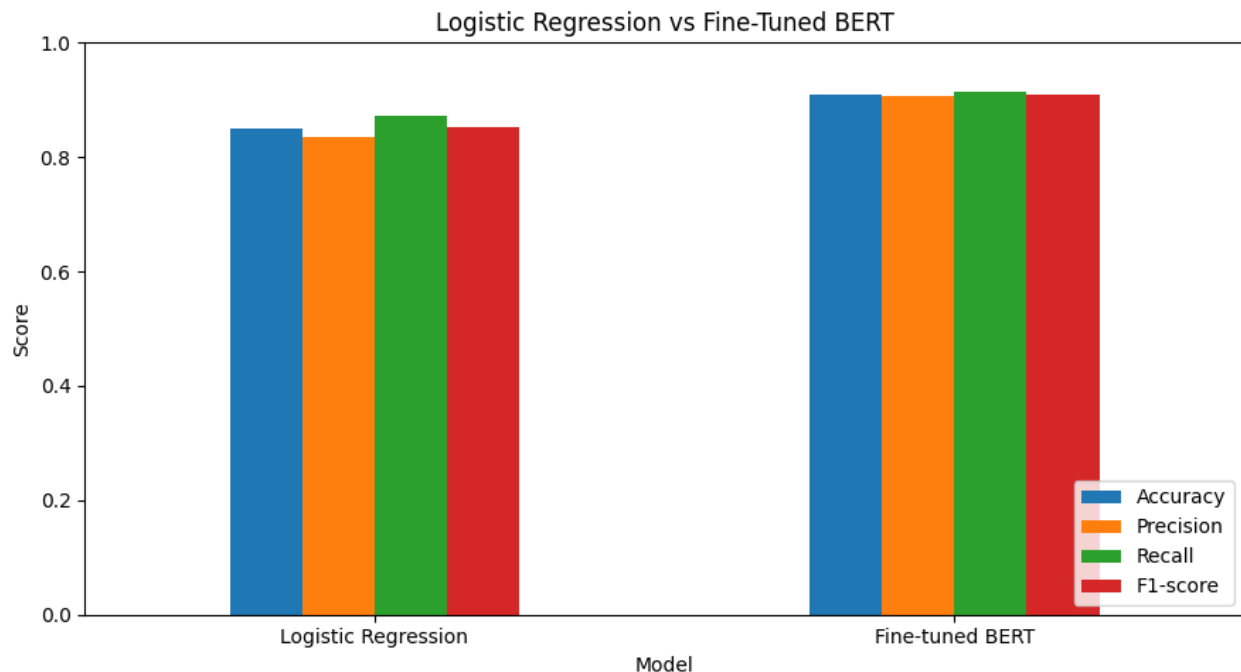
Logistic Regression VS Fine-Tuned BERT:					
	Model	Accuracy	Precision	Recall	F1-score
0	Logistic Regression	0.8505	0.8360	0.872	0.8536
1	Fine-tuned BERT	0.9095	0.9058	0.914	0.9099

As the results shows that both models have performed good at the sentiment classification task. Logistic Regression model shows a strong result and showed that even traditional machine learning algorithms still perform well in text classification problems. Fine-Tuned Transformer model outperformed in all evaluation metrics.

The performance of the Fine-Tuned Transformer model shows the ability to understand the context of the words instead of just considering their frequency. The model can accurately classify reviews which contain complex sentences as well as some words whose meaning is dependent on context. The TF-IDF algorithm considers words independently and cannot understand contextual relationships.

While the Fine-Tuned Transformer model has shown the best accuracy. It requires more computational resources and needs more time to train the model than Logistic Regression model.

Overall the results show that the fine-tuning of the pre-trained transformer model allows improving sentiment classification performance on the IMDB dataset.



Conclusion:

This assignment compares TF-IDF with Logistic Regression model and Fine-Tuned Transformer model for sentiment analysis for the IMDB dataset. Both models were good in classifying positive and negative reviews but the Fine-Tuned Transformer Model showed best results in terms of Accuracy, Precision, Recall, and F1-Score.

Even having faster training times and lower computational requirements the Logistic Regression model had less ability to understand the context behind natural language than Fine-Tuned Transformer Model. With the comparison of models shows that the transformer-based models perform well in sentiment analysis problems.